Venkata Lakshmi Parimala Pasupuleti

Staff Software Engineer · Backend Platform · Scalable Distributed Systems · Event-Driven Architecture · API Design · Reliability Engineering · Observability · Python · Redis · AWS

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Software Engineer with 8 years owning scalable distributed backend platform systems in production - foundational APIs, event-driven architecture, caching, and reliability tooling that product teams depend on to ship with confidence. At Entrata, led architecture and delivery of 20+ full-stack production platform modules at 99.5% uptime across 100K+ monthly active users: RabbitMQ event-driven processing (retry logic, idempotency, dead-letter handling), Redis caching for high-performance data access, Docker microservices redesign (50% scalability improvement), and observability tooling reducing incident triage time 40% across multi-tenant environments. Built AgentProbe - a backend platform for AI reliability with a CI-integrated regression gate, 5-metric observability engine, and high-throughput API layer. Mentoring 5+ engineers; 300+ code reviews. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Backend Platform: Python, SQL, FastAPI, PHP, JavaScript, TypeScript, Bash, Docker, Scalable Distributed Systems, Event-Driven Architecture, High-Throughput APIs, Low-Latency Systems, AWS (S3, EC2, RDS, Glue), Redis (caching), RabbitMQ (queues), REST APIs, CI/CD (GitHub Actions), Reliability Engineering, Observability, Incident Response, PostgreSQL, MySQL, Oracle, Linux, Git

AI & Agents: AI Integration, LangGraph, LangChain, Agentic AI, RAG Pipelines, LLM Orchestration, ChromaDB, Model Evaluation, Hallucination Detection, HITL, Prompt Engineering, OpenAI API, Anthropic API, MCP, FastAPI

ML & Data: Machine Learning, PyTorch, scikit-learn, NLP, BERT, PySpark, Feature Engineering, Statistical Modeling, Performance Optimization, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built scalable event-driven agentic backend platform (LangGraph stateful graph) - end-to-end architecture from data ingestion through conditional routing, caching, and validated output delivery; reliability guardrails blocking non-compliant outputs before production.
- Implemented RAG pipeline with observability on retrieval quality and response groundedness; performance-optimized for low-latency across distributed data sources.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built data-intensive ML pipelines (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$; full pipeline from ingestion through feature engineering, model optimization, and statistical validation across 190+ countries.
- Mentoring 600+ students in Python and database design; designed dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Led architecture and delivery of scalable backend platform services** - 20+ production modules at 99.5% uptime across 120+ Agile sprints; foundational API design and contract management across producers and consumers; 100K+ monthly active users with zero compliance violations across 7 years.
- **Designed and owned event-driven distributed systems** - RabbitMQ async message processing (retry logic, idempotency guarantees, dead-letter queues); Redis caching for high-performance low-latency data access; Docker microservices redesign (50% scalability improvement, 30% deployment reduction).
- **Built reliability and observability platform tooling** - internal incident response infrastructure reducing triage time 40%; performance monitoring across multi-tenant production environments; reliability engineering maintaining sub-10 bug count for 6+ consecutive months.
- **Performance optimization and cloud infrastructure at scale** - AWS S3, EC2, RDS; 200+ SQL query optimizations (30% performance gain) via execution plan analysis, indexing, and stored procedure refactoring; scalable data pipelines processing 180,000 domain records per release cycle; 100% PHI compliance.
- Mentoring 5+ engineers through design reviews and technical guidance; 300+ peer code reviews; established engineering standards across platform teams.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management backend platform modules (PHP, Python, PostgreSQL); scalable data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Backend Platform for AI Reliability & Observability

Platform · Reliability

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Scalable backend platform with 15-endpoint high-throughput FastAPI service, event-driven quality assessment engine (5 metrics: correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking production incidents before merge, LLM-as-judge scoring, and multi-provider routing for performance optimization. Full observability and incident response instrumentation across 1,240+ test cases.

LeaseI: Scalable Distributed Agentic Backend Platform

Backend · Distributed

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Distributed backend platform: 8-agent stateful graph with event-driven conditional routing, multi-source API data ingestion, caching, and reliability controls - HITL escalation when confidence thresholds not met. 93.6% precision, 87.9% F1 across 1,240 test cases; CI reliability gate with telemetry-driven quality scoring.

AI on FHIR: NLP-Driven Medical Records Query System

NLP · API

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Scalable NLP backend translating natural-language queries into FHIR API calls - reduced retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017